

Logical Functions And Lookup

Assignment

Question 1:

Write a formula to check if the marks in the cell b2 are greater than or equal to 40. If yes "Pass otherwise "fail".

Student name	marks
Riya	45
siya	72
raju	55
rahul	80
ram	30

Answer 1: Formula =IF(B2>=40,"pass","fail")

Output- Riya, siya,raju,rahul; **pass** and ram; **fail**.

Question 2:

A company gives a bonus if sales are above 40,000 and attendance is greater than or equal to 90%. Write a formula to calculate whether the employee will get a bonus or not.

Employee	sales	attendance
Arjun	55000	95
Simran	65000	85
Rohit	30000	90
Rohit Anjali	25000	65

Answer 2: Formula =IF(AND(B2>40000,C2>=90),"Bonus","No Bonus")

Output- Arjun; **Bonus**, simran,rohit,anjali; **no bonus**.

Question 3:

Write a nested IF formula: If marks in B2 "A" are $\geq 90 \rightarrow$, $\geq 75 \rightarrow$, $\geq 50 \rightarrow$, otherwise "B" "C" "Fail" .

Student name	marks
Raj	85
Neha	96
Vivek	57
Rani	92
Sneha	75

Answer: Formula =IF(>=90,"A",IF(B2>=75,"B",IF(B2>=50,"C","Fail")))

Output- raj,sneha; **B**, neha,rani; **A**, vivek; **C**.

Question 4:

Find the Department of Employee ID = E103. Also explain which LOOKUP function is used and why ?

EmpID	Name	Department
E101	Raju	IT
E102	Riya	HR
E103	Roshni	Finance
E104	Sejal	Sales

Answer: Formula =VLOOKUP("E103",A2:C5,3,FALSE)

Finance

VLOOKUP is used to;

- Find the employee ID in first column
- There is a need for value from the right-side column(department).

Question 5:

A company has set the following commission structure for its sales team:

Sales	Commission rate
0	5%
20000	10%
50000	15%

80000	20%
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If a salesperson achieves sales of ₹45,000, use the VLOOKUP function to determine the commission rate applicable.

Answer: Formula =VLOOKUP(45000,A2:B5,2,True) ;

The outcome will be 0.1 then go to the percentage sign in the ribbon(%) and it will give you the percentage value as 10%.

Question 6:

The HR department maintains the following leave records of employees:onus or not .

Employee name	Leaves taken	Leaves allowed
Alex	10	12
bob	10	15
Anna	8	12

Write an IF function to check whether an employee has Exceeded their allowed leave balance.

- If Leaves Taken > Allowed Leaves, display "Exceeded".
- Otherwise, display "OK".

Answer: Formula =IF(B2>C2,"exceeded","ok")

Outcome- bob, alex,anna; **OK**, as everyone has taken the leaves accordingly.

Question 7:

Find the salary of EmpID = E105 using INDEX-MATCH Function.

empID	Salary
E104	65000
E105	72000
E106	62000
E102	41000

Answer: Formula =INDEX(B16:B19,MATCH("E105",A16:A19))

Outcome- 72000

Question 8:

Use the XLOOKUP function to find the Department of the employee with Emp ID = E303.

EmpID	Department
E101	HR
E303	IT
E106	Sales
E301	Finance

Answer: Formula =XLOOKUP("E303",A23:A26,B23:B26)

Output- IT.

EmpID	name	department	Question 4
E101	raju	IT	Finance
E102	riya	HR	
E103	roshni	Finance	
E104	sejal	Sales	

Question 5

Sales	Commission rate	
0	5%	10%
20000	10%	
50000	15%	
80000	20%	

Question 7

emplID	Salary	
E104	65000	72000
E105	72000	
E106	62000	
E102	41000	

Question 8

EmpID	Department	
E101	HR	IT
E303	IT	
E106	Sales	
E301	Finance	